

Ignacio Peis

Technical University of Denmark, DTU
Dept. of Applied Mathematics and Computer Science
Section for Cognitive Systems
2800 Kgs. Lyngby
Denmark

ipeaz@dtu.dk
Building 321, 220
Website: <https://ipeis.github.io/>
Google Scholar

POSITIONS	Technical University of Denmark, DTU <i>Postdoctoral Researcher</i> Section for Cognitive Systems Dept. of Applied Mathematics and Computer Science (DTU Compute)	Oct. 2023 - Present
	Intor.AI <i>AI Researcher</i>	Nov. 2024 - Present
	EBIS Business School <i>Lecturer</i>	Oct. 2024 - Present
	Fundación UC3M & BBVA <i>Mentor at specialized courses</i>	Dec. 2020 - Dec. 2024
	Universidad Carlos III de Madrid <i>Predoctoral Researcher & Teaching Assistant</i> Signal Processing Group Supervisors: Prof. Antonio Artés Rodríguez Dr. Pablo Martínez Olmos	Oct. 2019 - Sept. 2023
	University of Cambridge <i>Visiting researcher</i> Machine Learning Group Supervisor: Prof. José Miguel Hernández-Lobato	Feb. 2021 - Feb. 2022
	Universidad Carlos III de Madrid <i>Research associate</i> Signal Processing Group Supervisor: Prof. Antonio Artés Rodríguez	Feb. 2017 - Sept. 2019
AFFILIATIONS	“La Caixa” Foundation <i>Postdoctoral Junior Leader Incoming</i>	Mar 2026 - Present
	Pioneer Centre for Artificial Intelligence <i>Postdoctoral Researcher</i>	Nov 2023 - Present
	Danish Data Science Academy (DDSA) <i>Postdoc Fellow</i>	June 2023 - Present

EDUCATION	Universidad Carlos III de Madrid <i>PhD in Probabilistic Machine Learning</i> <i>Advisors:</i> Prof. Antonio Artés Rodríguez Dr. Pablo Martínez Olmos <i>Cum Laude</i> <i>Outstanding PhD Thesis Award</i>	Oct 2018 - Oct 2023
	Universidad Carlos III de Madrid <i>M.Sc. in Telecommunications Engineering</i> <i>Dissertation with highest mark</i>	Sep. 2016 - Jul. 2018
	Universidad Carlos III de Madrid <i>M.Sc. in Multimedia and Communications</i> <i>Dissertation with highest mark</i> <i>One course with honors</i>	Sep. 2016 - Jul. 2018
	Universidad de Granada <i>B.Sc. in Telecommunications Engineering</i> <i>Five courses with honors</i>	Sep. 2012 - Jul. 2016
GRANTS	Junior Leader Incoming Postdoctoral Fellowship Title: <i>“Flow-Based Generative Meta-Learning for Reliable AI in Medicine”</i> Role: Principal Investigator Success rate: 5.93% Budget: 320.000,00 € Period: 2026-2029 Entity: <i>La Caixa Foundation</i> Country: Spain	
	DDSA Postdoctoral Fellowship Title: <i>“Implicit Neural Representations Generative Modelling”</i> Role: Principal Investigator Success rate: 11.32% Budget: DKK 1.200.000,00 (approx. 160.597,20 €) Period: 2023-2025 Entity: Danish Data Science Academy (DDSA) Novo Nordisk Foundation Country: Denmark	
	FPU Doctoral Fellowship Role: PhD Student Success rate: Period: 2019-2023 Entity: Ministry of Science, Innovation and Universities Country: Spain	
	Top Reviewer <i>International Conference on Learning Representations (ICLR)</i>	2026

Best Reviewer Award 2025
International Conference on Artificial Intelligence and Statistics (AISTATS)

Top Reviewer 2024
Neural Information Processing Systems (NeurIPS)

Top-10% Reviewer 2023
International Conference on Artificial Intelligence and Statistics (AISTATS)

Outstanding PhD Thesis Award 2023
Universidad Carlos III de Madrid
 PhD Defense Committee:

Dr. Daniel Hernández Lobato, UAM
 Dr. Francisco R. Ruiz, Google Deepmind
 Prof. Fernando Pérez Cruz, ETH, SDSC

CONFERENCE PAPERS Alberte Heering, **Ignacio Peis** and Jes Frellsen. Latent Diffusion for Missing Data. *Symposium on Probabilistic Machine Learning (ProbML)*, 2026.
 [pdf]

Noa Margeta, Jes Frellsen and **Ignacio Peis**. Denoising Uncertainty via Gaussian Distributional Diffusion Models *Symposium on Probabilistic Machine Learning (ProbML)*, 2026.
 [pdf]

Ignacio Peis, Batuhan Koyuncu, Isabel Valera and Jes Frellsen. Hyper-Transforming Latent Diffusion Models. *Proceedings of the 42nd International Conference on Machine Learning, in Proceedings of Machine Learning Research*, 2025.
 [pdf] [poster] [video] [slides] [code]

Batuhan Koyuncu, Pablo Sánchez-Martín, **Ignacio Peis**, Pablo M. Olmos and Isabel Valera. Variational Mixture of HyperGenerators for Learning Distributions Over Functions. *Proceedings of the 40th International Conference on Machine Learning, in Proceedings of Machine Learning Research 202:17660-17683*, 2023.
 [pdf] [poster] [video] [code]

Ignacio Peis, Chao Ma and José Miguel Hernández-Lobato. Missing Data Imputation and Acquisition with Deep Hierarchical Models and Hamiltonian Monte Carlo. In *Advances in Neural Information Processing Systems (NeurIPS) 35: 35839-35851*, 2022.
 [pdf] [code] [slides] [video] [poster]

Ignacio Peis, Ignacio A. Illán, Francisco J. Martínez-Murcia, Fermín Segovia, Juan M. Górriz, Javier Ramírez, E. W. Lang & Diego Salas-Gonzalez. MRI brain segmentation using hidden Markov random fields with alpha-stable distributions. In *IEEE Nuclear Science Symposium, Medical Imaging Conference and Room-Temperature Semiconductor Detector Workshop (NSS/MIC/RTSD)* (pp. 1-3). IEEE, 2016.
 [html]

JOURNAL PAPERS	Berian James, Stefan Pollok, Ignacio Peis , Jes Frellsen and Rasmus Bjørk. Scalable physical source-to-field inference with hypernetworks. <i>Transactions of Machine Learning Research</i> , 2835-8856, 2026. [pdf] [code]	
	Ignacio Peis , Pablo M. Olmos and Antonio Artés-Rodríguez. Unsupervised Learning of Global Factors in Deep Generative Models. In <i>Pattern Recognition</i> , vol. 134, p. 109130, 2022. [pdf] [code]	
	Ignacio Peis , Javier-David López-Moríñigo, M. Mercedes Pérez-Rodríguez, Maria-Luisa Barrigón, Marta Ruiz-Gómez, Antonio Artés-Rodríguez and Enrique Baca-García. Actigraphic recording of motor activity in depressed inpatients: a novel computational approach to prediction of clinical course and hospital discharge. In <i>Scientific reports</i> , 10. Nature, 2020. [pdf]	
	Ignacio Peis , Pablo M. Olmos, Constanza Vera-Varela, Maria-Luisa Barrigón, Philippe Courtet, Enrique Baca-García and Antonio Artés-Rodríguez. Deep Sequential Models for Suicidal Ideation from Multiple Source Data. In <i>Journal of Biomedical and Health Informatics</i> , vol. 23, no. 6. IEEE, 2019. [pdf] [html]	
	D. Castillo-Barnes, I. Peis , F. J. Martínez-Murcia, F. Segovia, I. A. Illán, J. M. Górriz, J. Ramírez, D. Salas-Gonzalez. A Heavy Tailed Expectation Maximization Hidden Markov Random Field Model with Applications to Segmentation of MRI . In <i>Frontiers in Neuroinformatics</i> , 11, 66, 2017. [pdf]	
PREPRINTS	Ignacio Peis , Pablo M. Olmos and Jes Frellsen. Order-Agnostic Autoregressive Modelling with Missing Data. <i>arXiv preprint arXiv:2605.06355</i> , 2026. [pdf]	
TEACHING	<i>Data Mining, Machine Learning and Deep Learning</i> M.Sc. in Data Science and Artificial Intelligence EBIS, EUNEIZ	2024 - Present
	<i>Generative AI</i> Specialized Course BBVA, Fundación UC3M	2025 - Present
	<i>Natural Language Processing</i> Specialized Course BBVA, Fundación UC3M	2026 - Present
	<i>Advanced Machine Learning</i> Specialized Course BBVA, Fundación UC3M	2023 - Present

	<i>Intermediate Machine Learning and Feature Engineering</i> Specialized Course BBVA, Fundación UC3M	2022 - Present
	<i>Machine Learning Fundamentals</i> BBVA, Fundación UC3M	2020 - Present
	<i>Machine Learning II</i> Bachelor in Data Science and Engineering Teaching evaluation survey: 4.67/5 (2020), 4.63/5 (2022) Universidad Carlos III de Madrid	2020, 2022
	<i>Neural Networks</i> Bachelor in Data Science and Engineering Teaching evaluation survey: 4.13/5 Universidad Carlos III de Madrid	2022
	<i>Communications Theory</i> Bachelor in Telecommunications Engineering Bachelor in Sound and Image Engineering Teaching evaluation survey: 3.94/5, 4.14/5 Universidad Carlos III de Madrid	2022
SUPERVISION	Master Students at Technical University of Denmark Noa Margeta (2025), Nikolaj Bach Meineche (2025), Eva Kastelan (2026), Frederik Emil Nagel (2026)	2025 - 2026
	Visting PhD Students at Technical University of Denmark Francisco Javier Sáez Maldonado (Universidad de Granada, 2026)	2026
SERVICE	Generative Modelling Summer School (GeMSS) Local Organizer	2027
	Andaluz.IA Conference Scientific Commitee	2025 - Present
TALKS	<i>Hyper-Transforming Latent Diffusion Models</i> Andaluz.IA forum [slides]	Dec. 2025
	<i>Hyper-Transforming Latent Diffusion Models</i> Signal Processing Group Universidad Carlos III de Madrid	Dec. 2025
	<i>Hyper-Transforming Latent Diffusion Models</i> Section for Cognitive Systems [slides]	Jun. 2025

	<i>Hyper-Transforming Latent Diffusion Models</i> CITIC, Universidad de Granada [slides]	Apr. 2025
	<i>Variational Mixture of HyperGenerators for Learning Distributions Over Functions</i> Andaluz.IA forum [slides]	Dec. 2023
	<i>Information Acquisition and Distributions of Functions with Deep Generative Models</i> Pioneer Centre for Artificial Intelligence, Copenhagen, Denmark [slides]	Jun. 2023
	<i>Missing Data Imputation and Acquisition with Deep Hierarchical Models and Hamiltonian Monte Carlo</i> Oral (video) and poster presentation NeurIPS22, New Orleans, USA [slides] [video] [poster]	Dec. 2022
	<i>Missing Data Imputation and Acquisition with Deep Hierarchical Models and Hamiltonian Monte Carlo</i> Signal Processing Group Universidad Carlos III de Madrid	Jun. 2022
	<i>Deep Sequential Models for Suicidal Ideation from Multiple Source Data</i> Signal Processing Group Universidad Carlos III de Madrid	Jul. 2018
REVIEWING	The European Conference on Artificial Intelligence (ECAI) Workshop on Weight-Space Learning (ICLR) International Conference on Machine Learning (ICML) Transactions on Machine Learning Research (TMLR) Northern Lights Deep Learning Conference (NLDL) Workshop Proposals Reviewer (EurIPS) International Conference on Learning Representations (ICLR) Top Reviewer (2026) Neural Information Processing Systems (NeurIPS) Top Reviewer (2024) Association for Uncertainty in Artificial Intelligence (UAI) Artificial Intelligence and Statistics (AISTATS) Top-10% Reviewer (2023) Best Reviewer Award (2025) Journal of Machine Learning Research (JMLR) Journal of Biomedical and Health Informatics (JBHI) Workshop on Deep Generative Models for Health (DGM4H)	2025 - Present 2025 - Present 2025 - Present 2025 - Present 2025 - Present 2025 2024 - Present 2024 - Present 2024 - Present 2022 - Present 2022 - Present 2022 - Present 2018 - Present 2023

PROJECTS	Title: PRACTICO-CM: “<i>Psiquiatría Computacional y Modelos Integrales de Comportamiento</i>”	
	Role: Research team member	
	Budget: 645.775,90 €	
	Period: 2019 - 2022	
	Entity: Consejería de Educación e Investigación, CAM	
	Country: Spain	
	Title: “<i>Aprendizaje máquina y computación masiva para medicina personalizada y análisis cuantitativo del clima</i>”	
	Role: Research team member	
	Budget: 124.509,00 €	
	Period: 2019 - 2022	
	Entity: Agencia Estatal de Investigación (AEI)	
	Country: Spain	
	Title: “<i>Métodos computacionales bayesianos avanzados para estimación, predicción y control en sistemas multisensoriales complejos</i>”	
	Role: Research team member	
	Budget: 314.600,00 €	
	Period: 2016 - 2029	
	Entity: Ministerio de Economía, Comercio y Empresa	
	Country: Spain	
	Title: “<i>Modelos profundos y explicables basados en variables latentes para salud mental</i>”	
	Role: Research team member	
	Budget: 190.212,00 €	
	Period: 2022 - 2025	
	Entity: Agencia Estatal de Investigación (AEI)	
	Country: Spain	
COURSES	Model Parallelism: Building and Deploying Large Neural Networks	Oct. 2024
	<i>NVIDIA</i>	
	AI and Machine Learning in Healthcare Summer School	Sep. 2022
	<i>Cambridge Centre for AI in Medicine</i>	
	<i>University of Cambridge, UK</i>	
	Gaussian Process and Uncertainty Quantification Summer School, 2020.	Sep. 2020
	<i>University of Sheffield, UK</i>	
	Machine Learning Summer School (MLSS)	Sep. 2019
	<i>Skoltech Institute of Science and Technology, Moscow, Russia</i>	
	Machine Learning Frontiers in Precision Medicine (MLFPM) 1st Summer School	Sep. 2019

ETH Zurich, Switzerland

Machine Learning Summer School (MLSS)

Sep. 2018

Universidad Autónoma de Madrid, Spain

Gaussian Process and Uncertainty Quantification Summer School, 2017.

Sep. 2017

University of Sheffield, UK

DISSERTATIONS

Ignacio Peis. Advanced Inference and Representation Learning Methods in Variational Autoencoders. ***PhD Thesis Dissertation*** (*Probabilistic Machine Learning*), 2023.
[pdf] [slides]

Ignacio Peis. Deep sequential models with attention for psychiatric patients clinical assessment. ***M.Sc. Thesis Dissertation*** (*Multimedia and Communications*), 2018.

Ignacio Peis. Activity monitoring in depressed patients in the hospital setting: a pilot study testing new methods of actigraphy data analysis for predicting clinical progress and date of hospital discharge. ***M.Sc. Thesis Dissertation*** (*Telecommunications Engineering*), 2018.

Ignacio Peis. Hidden Markov Random Fields with alpha-stable distributions for brain Magnetic Resonance Images. ***B.Sc. Thesis Dissertation*** (*Telecommunications Engineering*), 2016.